

Boolean Values, conditions, algebra

Conditions, AND, OR, NOT, if statements



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Topics list

1. Boolean values
2. Boolean conditions and variables
3. Compound conditions / Boolean Algebra
 - i. Logical operators
 - ii. Bringing conditions together

Boolean values

- A **boolean** is a data type that can only have two possible values:
 - true
 - false
- Think of it like a light switch - it's either ON or OFF. There's no in-between!
- Booleans are named after George Boole, a mathematician who developed Boolean algebra.

Why do we need Booleans?

- In programming, we constantly need to make decisions:
 - Is the user old enough to vote?
 - Does the student have enough credits to graduate?
 - Is there enough money in the account for this purchase?
 - Has the password been entered correctly?
- All of these questions have
 - YES/NO or
 - TRUE/FALSE answers –
- that's where booleans come in!

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Boolean Conditions

- A boolean condition is an expression that evaluates to either
 - true or
 - false.
- Example Boolean Conditions:

```
age >= 18           // Is age greater than or equal to 18?  
temperature < 0     // Is temperature less than 0?  
score == 100        // Is score equal to 100?  
balance != 0        // Is balance not equal to 0?
```

Java Relational Operators

Operator	Use	Returns true if...
>	op1 > op2	op1 is greater than op2
>=	op1 >= op2	op1 is greater than or equal to op2
<	op1 < op2	op1 is less than to op2
<=	op1 <= op2	op1 is less than or equal to op2
==	op1 == op2	op1 and op2 are equal
!=	op1 != op2	op1 and op2 are not equal

BEWARE = is an assignment operator.

It doesn't test for equality. Use == to test for equality in primitive types

Source: http://www.freejavaguide.com/relational_operators.htm

Boolean Variables - use in if statement

- A boolean variable stores a true or false value.
- They can be used to make your code more readable and lets you reuse conditions.

```
boolean canVote;    // Is age greater than or equal to 18?  
  
canVote = age >= 18;  
  
if (canVote) System.out.println("You can vote!");
```

```
boolean found = false; //checking are vars equal to a lookingFor val  
    :    // some code  
if (val == lookingFor) {  
    found = true;  
}
```

Simple conditions – one condition only

```
if (age >= 18 ) { // Is age greater than 18
    System.out.println("I'm an adult!");
}
```

```
if (price < 0) { // invalid price
    System.out.println("Too cheap!");
}
```

Topics list

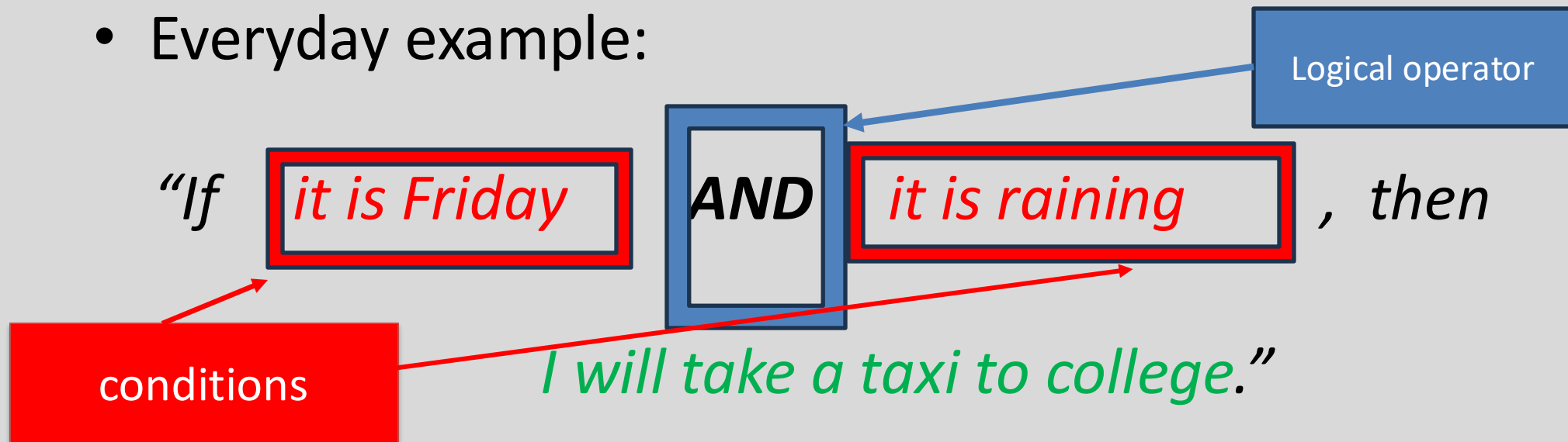
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Compound conditions – using Boolean algebra

- When we use a combination of conditions we call it a compound condition.
- This is done by taking simple compound statements and putting it with another condition with either an
 - and (&&) or
 - or(||)
 - not (!)

Boolean Conditions with AND (&&)

- An AND condition requires **both tests** to be true.
- Everyday example:



The two conditions are brought together to make one condition.

Boolean Conditions with AND (&&)

*“If **it is Friday** **AND** **it is raining** , then
I will take a taxi to college.”*

Only if both conditions are true will I take a taxi.

I will **NOT** take a taxi if either (or both) of the following is **False**

‘it is Friday’ is false (e.g. it’s Thursday)

‘it is raining’ is false (i.e. it is dry)

Code example using AND (&&)

```
if ( (mark >= 0 ) && ( mark <= 100) ){  
    System.out.println("This is a valid % mark.");  
}
```

Question : Will the following values of mark lead the condition to be true or false:

mark has value 0 ?

 -1 ?

 100 ?

 101 ?

Boolean Conditions with OR (||)

*“If **it is Friday** **OR** **it is raining** , then
I will take a taxi to college.”*

if either or both conditions are true will I take a taxi.

The only way I won't take a taxi is if both of the conditions are false.

Code example using AND (&&)

```
if ( (mark < 0 ) || ( mark > 100) ){  
    System.out.println("This is NOT a valid % mark.");  
}
```

Question : Will the following values of mark lead the condition to be true or false:

mark has value 0 ?

 -1 ?

 100 ?

 101 ?

Boolean Conditions with NOT (!)

*“If NOT(it is Friday) , then
I will take the bike to college.”*

If it is NOT Friday I will
take the bike to college

NOT True is False

! True → False

NOT False is True

! False -> True

Code example using NOT (!)

```
boolean isEligible = false;  
if ( !isEligible ){  
    System.out.println("Sorry you cannot vote.");  
}
```

Question : Will the following values of ineligible lead to "Sorry you cannot vote." being printed:

ineligible has value true ?
 false ?

Questions?

